

DL Graded Quiz M6

Question 1

Consider a set of N assets with expected returns μ and variance-covariance matrix Σ , as in Lesson 4. The portfolio $w = (w_1, \dots, w_N)^T$ with the maximum Sharpe ratio is given by which of the following?

- ☐ $w^* = \frac{\Sigma \mu}{\mu^T \Sigma \mu}$
- ☐ $w^* = \frac{\mu^T \Sigma^{-1} \mu}{\mu^T \Sigma^{-1} \mu}$
- ☐ $w^* = \frac{\Sigma^{-1} \mu}{\mu^T \Sigma^{-1} \mu}$
- ☒ $w^* = \Sigma^{-1} \mu$

Question 2

Consider the time series of two random variables (x_1, x_2) . Over a common time period, we observe the time series of realizations $x_1 = (0.1, -0.25, 0.75, 1)$ and $x_2 = (0.421, 0.25, 1.75, 1.15)$. The first principal component of these time series explains the following percentage of the variation in the data:

- ☐ 87.41%
- ☐ 92.66%
- ☐ 93.06%
- ☒ 88.91%

Question 3

Choose which of the following constraints may be useful in autoencoders. (Choose all that apply.)

- ☒ Choosing encodings that have the maximum degree of correlation
- ☒ Choosing weights in each layer such that $W^T W = Identity$
- ☒ Tying the activations of the output layer with those of the bottleneck
- ☒ Setting orthonormal weights

Question 4

Determine the **wrong** answer about autoencoders.

- ☐ They represent a generalization of principal component analysis that allows for non-linearities.
- ☐ They are explicitly designed to obtain predictions in classification problems, where the output is the label associated with some features.
- ☐ Their architecture has two parts: an encoder and a decoder.
- ☒ The bottleneck must have a smaller dimension than the input data.

Question 5

In the equation $R_t = \mathbf{a} + \mathbf{B}\mathbf{F}_t + \mathbf{e}_t$, the term $\mathbf{a}$ represents which of the following?

- ☐ The bias terms associated with the output layer of an autoencoder with linear activation
- ☐ The bias terms associated with the input layer of an autoencoder with linear activation
- ☐ The bias terms associated with the bottleneck layer of an autoencoder with linear activation
- ☒ The bias terms associated with the input layer of an autoencoder with non-linear activation

Question 6

Consider the time series of two random variables (x_1, x_2) . Over a common time period, we observe the time series of realizations $x_1 = (0.1, -0.25, 0.75, 1)$ and $x_2 = (0.421, -0.25, 1.75, 0.96)$. The first principal component of these time series explains the following percentage of the variation in the data:

- ☐ 93.06%
- ☐ 92.66%
- ☐ 87.41%
- ☒ 88.91%

Question 7

Choose which of the following constraints may be useful in autoencoders. (Choose all that apply.)

- ☒ Setting weights with unit norm

- ☒ An L1 penalty on the dimension of the bottleneck
- ☒ Setting orthonormal encodings
- ☒ Choosing encodings that have the maximum degree of correlation

Question 8

Determine the **wrong** answer in association with principal component analysis.

- ☐ It can be represented as a linear autoencoder with a single layer.
- ☐ The latent encodings have unit variance and are uncorrelated.
- ☐ It minimizes the reconstruction error subject to orthonormality of the principal directions.
- ☒ The principal directions are given by orthogonal eigenvectors of the variance covariance matrix of the input data.

Question 9

Consider a set of N assets with expected returns μ and variance-covariance matrix Σ , as in Lesson 4. The expected return of a portfolio $w = (w_1, \dots, w_2)^T$ is given by which of the following?

- ☐ μw
- ☐ $w^T \mu w$
- ☐ $w \mu$
- ☒ $w^T \mu$



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QUESTIONS

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|----|----|----|----|
| 1 | 2 | 3 | 4 |
| 5 | 6 | 7 | 8 |
| 9 | 10 | 11 | 12 |
| 13 | 14 | 15 | 16 |

Question 10

Choose the **right** answer regarding the loadings generated by the application of autoencoders to stock returns as in Lesson 2. (Choose all that apply.)

- ☒ They capture the exposure of each stock to each latent source of comovement (encoding).
- ☒ They capture the eigenvalues of the variance covariance matrix of the data.
- ☒ They are extracted from the weights of the output layer.
- ☒ They represent the latent factors that generate comovement across stocks.

Question 11

Choose the **wrong** statement associated with sparse autoencoders. (Choose all that apply.)

- ☒ They impose sparsity on the bottleneck's activations.

- ✓ They impose orthogonality of the bottleneck weights.
- ✓ The loss function includes a penalty term $\lambda_1 \|\hat{X}\|_1$.
- ✓ They set an L2 penalty on the bottleneck's activations.

Question 12

Consider a single layer linear autoencoder with bottleneck dimension equal to k . Let λ_i denote the eigenvalues of the variance covariance matrix of the data, $i = 1, \dots, n$, with $k < n$. Determine the right answer:

- The lower bound on the MSE of the autoencoder is given by $\sum_{i=1}^k \lambda_i$ in the training sample.
- The lower bound on the MSE of the autoencoder is given by $\sum_{i=k+1}^n \lambda_i$ in the training sample.
- It is not possible to know the lower bound on the MSE in the training sample.
- The lower bound on the MSE of the autoencoder is given by $\sum_{i=1}^n \lambda_i$ in the training sample.

Question 13

Consider a set of N assets with expected returns μ and variance-covariance matrix Σ , as in Lesson 4. The variance of the returns of a portfolio $w = (w_1, \dots, w_N)^T$ is given by which of the following?

- $w \Sigma w^T$
- $w \Sigma^{-1} w^T$

☐ $w^T \Sigma w$

☒ $w \Sigma^{-1}$

Question 14

Determine which of the following statements **cannot** be true. (Choose all that apply.)

- ☒ "We trained an undercomplete nonlinear autoencoder with a bottleneck dimension of 2 whose MSE reconstruction error in the training sample fell below the MSE of PCA with 2 components."
- ☒ "We trained a linear autoencoder with a bottleneck dimension greater than the dimension of the input data."
- ☒ "Including an additional component in PCA increased the total variance explained from 10% to 85%."
- ☒ "We trained an undercomplete linear autoencoder with a bottleneck dimension of 1 whose MSE reconstruction error in the training sample fell below the MSE of PCA with 2 components."

Question 15

Setting constraints may be useful in autoencoders for which of the following reasons? (Choose all that apply)

- ☒ They alleviate exploding/vanishing gradient problems.

- ☒ They can generate correlation between the encodings.
- ☒ They shrink the values of the trainable parameters equivalent to an L1 penalty.
- ☒ They represent a regularization tool.

Question 16

Complete the stem:

The orthonormal eigenvectors of the variance covariance matrix of the input data _____.

- ☐ are ordered from highest to lowest
- ☐ generate correlated encodings
- ☐ are a unique solution to the MSE-minimization problem
- ☒ minimize the reconstruction error, measured with the MSE

SUBMIT